



RARE EARTH VS FERRITE (RARE-EARTH-FREE) MOTORS

Category	Rare Earth Motors	Ferrite/Magnet-Free Motors	Startup Insights
Cost	• Priced at ₹10,000-₹12,000 per kg • Prices fluctuate with global demand and export policy	• Priced around ₹500-₹1000 per kg • Lower and stable cost due to abundant raw material	Conifer: Manufacturing process is up to 90% cheaper; uses less copper and steel, no tooling
Efficiency	• Highest torque density, essential for premium EVs and aerospace • Typical efficiency: IE4 to IE5	• Slightly lower torque, but new axial-flux and reluctance-based topologies deliver IE5+ efficiency	• Conifer: Ferrite motors increase range by ~30% and reduce size by ~15% vs hub motors • Chara: Synchronous-reluctance motors achieve PMSM-level torque using rotor geometry + software • Viridian Igni: Matches PMSM efficiency within 15-20% while reducing dependency and emissions
Reliability and Thermal Performance	• Magnets lose strength above 150°C • Require protective coatings to prevent corrosion	• Heat-stable up to 180°C • Naturally corrosion-resistant and suited for India's hot, dusty climate	Chara: Motors built for 42°C ambient with IP67-rated enclosures and tested for vibration and shock
Scalability and Localisation	• India has limited mining and refining capacity • High capital investment required for magnet manufacturing	• Can be fully localised with existing Indian supply chains • Compatible with current motor manufacturing infrastructure	All three (Chara, Conifer, Viridian Igni): Ferrite and magnet-free designs enable domestic sourcing, faster scale-up, and lower risk

Historically, India's focus had been on thorium and uranium extraction, but "now rare earths have become equally important." According to Sood, IREL has developed the "full capability to extract both light and heavy rare earth oxides, including those required for making rare earth permanent magnets such as neodymium and praseodymium." **India currently produces around 1500 tons of rare earth oxides annually**, marking a crucial step toward domestic self-reliance.

Key components of India's rare earth strategy include:

- **Developing a complete technology chain.** From oxide extraction to metal conversion and alloying with elements like iron and boron to produce rare earth magnets. "This complete technology chain has now been developed domestically," Sood noted, adding that several Indian companies are preparing to manufacture magnets using indigenously sourced materials.

- **Investing in rare-earth-free permanent magnets**, which do not use rare earth elements but can still achieve high magnetisation suitable for various applications, including automobiles. These magnets, though slightly larger, are functional and cost-effective alternatives that help reduce import dependence.

Highlighting these initiatives, Sood concluded, "The effort is well underway to make India technologically self-reliant and globally competitive in this strategically important field."

Now, let us examine India's policy framework in this regard. The strategy has shifted from restrictive atomic-era controls to a forward-looking industrial policy. In 1962, the Atomic Energy Act kept thorium- and monazite-rich sands under state control, ensuring security but stifling private exploration.

Reforms accelerated with the Mines and Minerals (Development and Regulation) Amendment Act, 2023, which streamlined auctions for 24 strategic minerals and opened exploration licences for 29 others, inviting private and foreign investment. The Geological Survey of India (GSI) and Atomic Minerals Directorate (AMD) now have 195 exploration projects underway, with 227 more planned for FY 2025-26.

A milestone came this year, when the National Critical Minerals Mission (NCMM) was launched in January 2025 with a ₹163 billion outlay. It aims to unify exploration, recycling, research, and industry collaboration. It targets 1200 exploration projects, ₹15 billion in recycling incentives and 1000 patents by FY 2030-31, with strong R&D support

through the National Mineral Exploration Trust.

On the manufacturing front, a ₹73 billion scheme for rare earth magnets offering up to 50% cost subsidies seeks to produce 6000 metric tonnes annually by 2030. Complementary tariff exemptions on critical inputs and plans for a national stockpile underscore India's push to build a secure and self-reliant critical minerals ecosystem.

At present, India imports around 80% of its rare earth elements from China, prompting the creation of the NCMS and efforts to diversify to countries such as Kazakhstan. As Suraj Rengarajan, Managing Director of Applied Materials' Semiconductor Products Group, notes, "Every motor, for example, needs neodymium (a rare earth metal), and obtaining that is becoming more critical and increasingly difficult."

Here, one question remains unanswered: Is the government merely fortifying access to raw materials, or genuinely pursuing alternatives that could reduce dependence altogether?

So far, most policy attention has centred on supply-chain resilience, new exploration blocks, magnet manufacturing incentives, and recycling schemes. These are necessary moves, but they remain defensive.